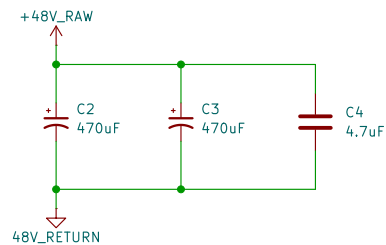
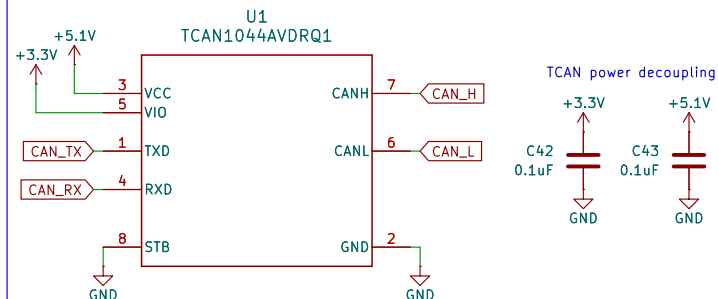


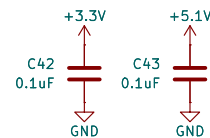
Tether 48V Decoupling Caps



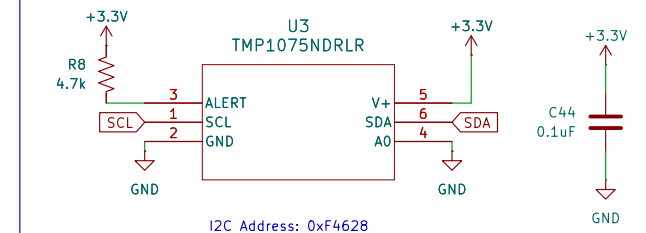
CAN Transceiver



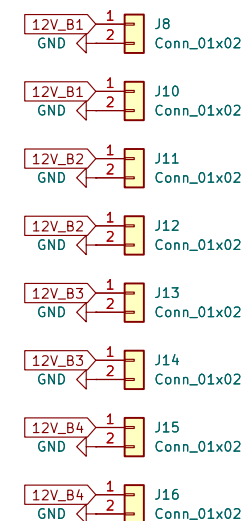
TCAN power decoupling



Temp Sensor



XT60 ESC Power



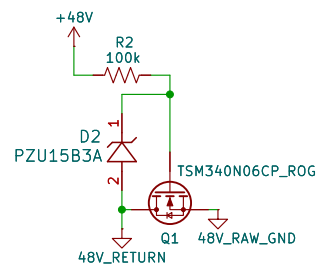
[WARNING]!

48V with 30A fuse gives 1440W.
4x360W bricks takes 1440W potentially
That on top of 50W brick means at max
load the fuse will blow. Unlikely but
keep in mind.

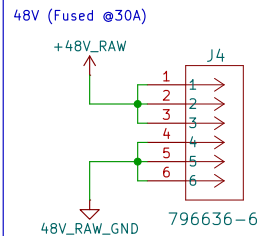
[WARNING]!

ALWAYS PLUG TETHER IN FIRST
BEFORE POWERING ON
No inrush / precharge is implemented

Reverse Polarity Protection



Tether Wall Connector



Power



File: power.kicad_sch

MCU



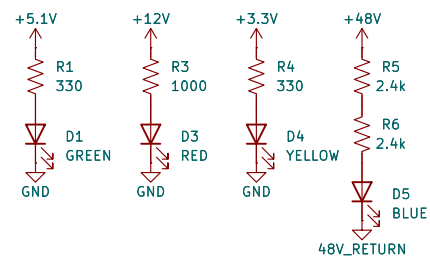
File: STM32MCU.kicad_sch

Power Monitoring

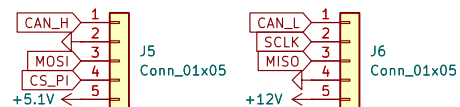


File: pm.kicad_sch

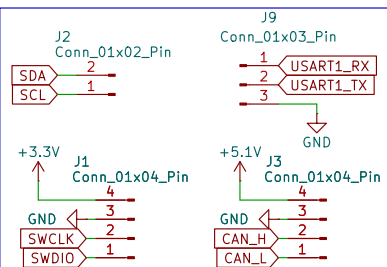
Voltage LEDs



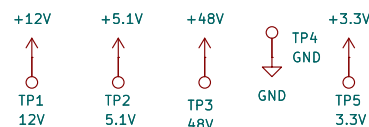
Board Output Connectors



Debugging Pins



Test Points



Purdue ROV

Sheet: /

File: X19-Power-Slab-Board.kicad_sch

Title: Purdue ROV Subsystem Board

Size: A4

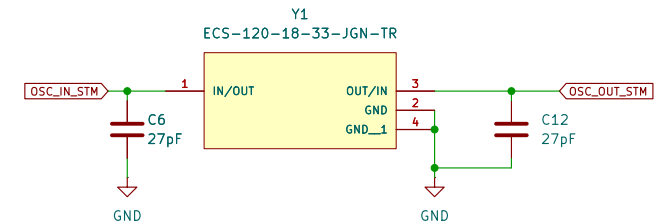
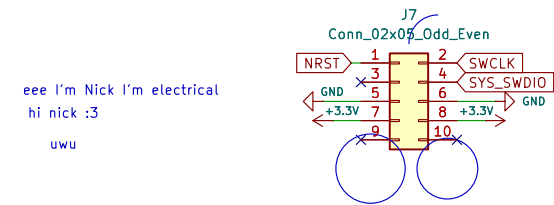
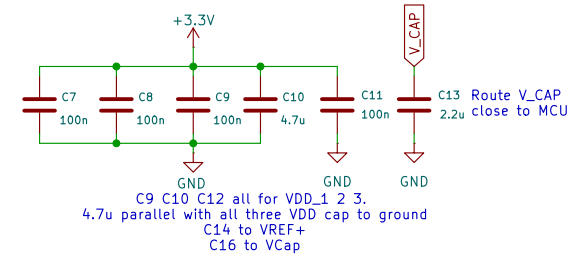
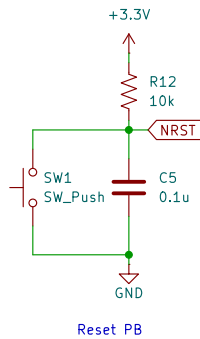
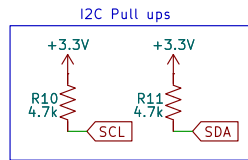
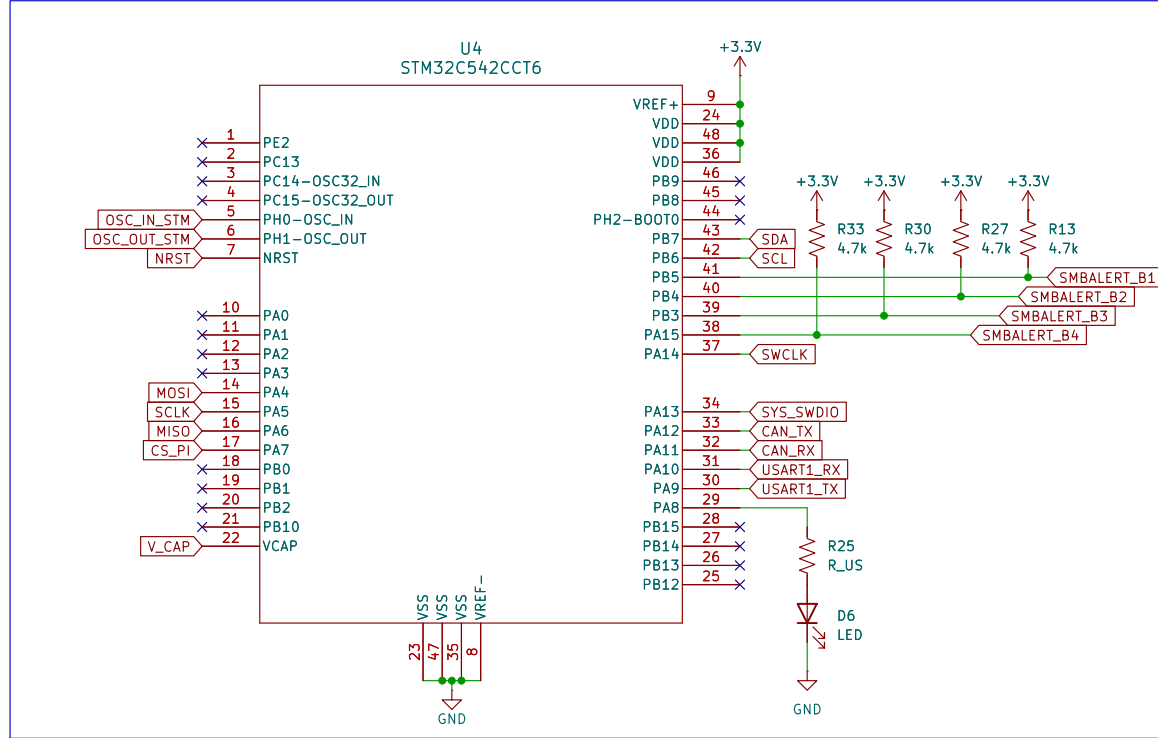
Date:

KiCad E.D.A. 10.0.5-unknown

Rev:

Id: 1/4

STM32 MCU Pinout



Sheet: /MCU/
File: STM32MCU.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.5-unknown

Rev:

Id: 2/4

The diagram shows the MIC5504-3.3YM5-TR voltage regulator. The input side (left) features a +5.1V supply connected to pin 1 (VIN) through a 1uF capacitor (C15). Pin 2 (GND) is connected to ground. The output side (right) shows the output voltage (VOUT) at pin 5, which is +3.3V, connected to a 1uF capacitor (C19) to ground. Pin 4 (NC) is marked with an 'X'.

INPUT

Low ESR 100V Electrolytic (VIN+ / VIN-)

Low ESR 100V (VIN+ / VIN-)

OUTPUT

Low ESR 25V (12V_B# / GND)

100V or 25V??? (12V_B# / GND)

B5 PKU5511ES1

INPUT

Low ESR Input
100V Rated

+48V

100V XTR

C14 100uF

C16 1uF

48V_RETURN

OUTPUT

10V XTR

+5.1V_UNFILT

C17 1uF

C18 470uF

Use 25V Rated Cap

GND

Datasheet specifies
>330uF output capacitance
between OUT+, OUT-

Resistor Values chosen
in datasheet pg.15
USE 1% TOLERANCE RESISTORS

SMBALERT pull up is on MCU sheet

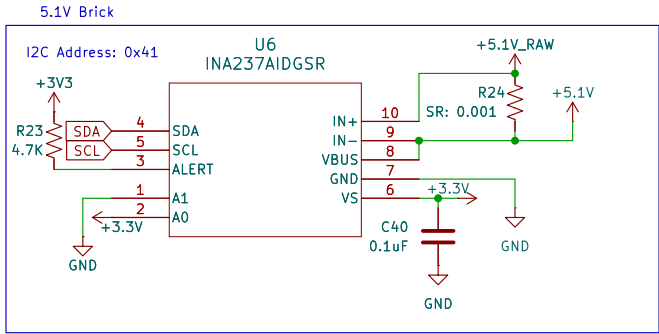
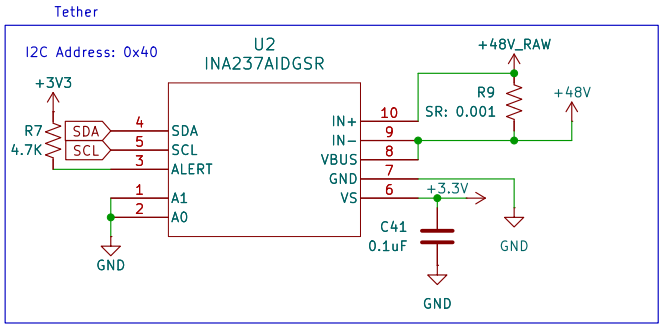
12C Address: 0x20 (Hex)

12C Address: 0x21 (Hex)

12C Address: 0x22 (Hex)

12C Address: 0x23 (Hex)

Title:		
Size: A4	Date:	Rev:
KtCad E.D.A. 10.0.5-unknown		Id: 3/4



Sheet: /Power Monitoring/
File: pm.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.5-unknown

Rev:

Id: 4/4